

模拟洗牌发牌

复习

- | | |
|-------------|----|
| 1. 位运算概述 | 了解 |
| 2. 按位取反 | 重点 |
| 3. 按位与 | 重点 |
| 4. 按位或 | 重点 |
| 5. 按位异或 | 重点 |
| 6. 左移和右移 | 重点 |
| 7. 位运算赋值运算符 | 熟悉 |

目录

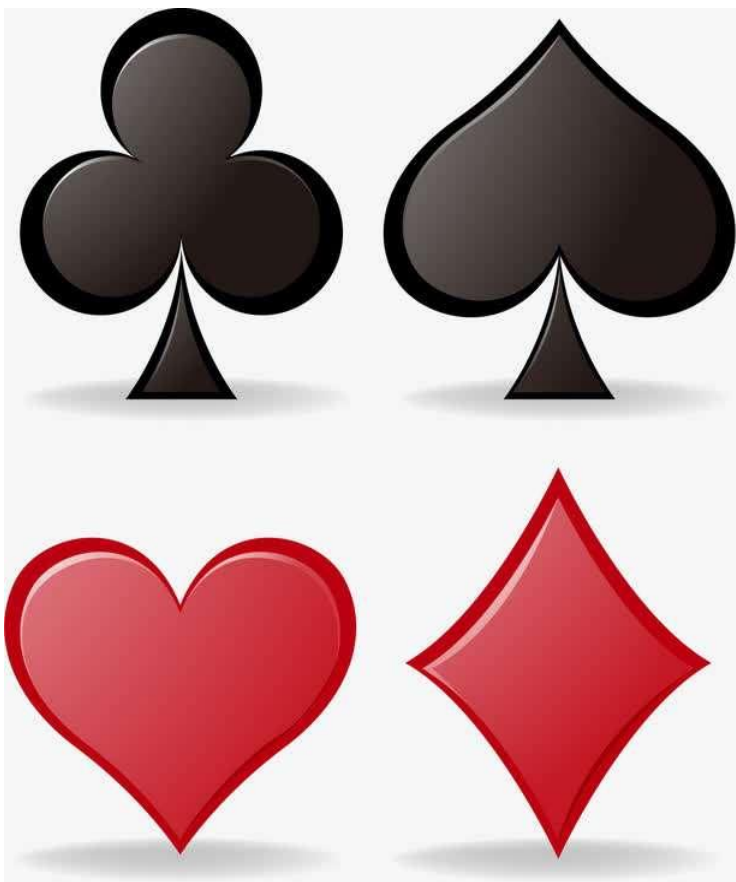
1. 模拟洗牌发牌概述
2. 生成54张牌
3. 洗牌
4. 模拟发牌

模拟洗牌发牌概述

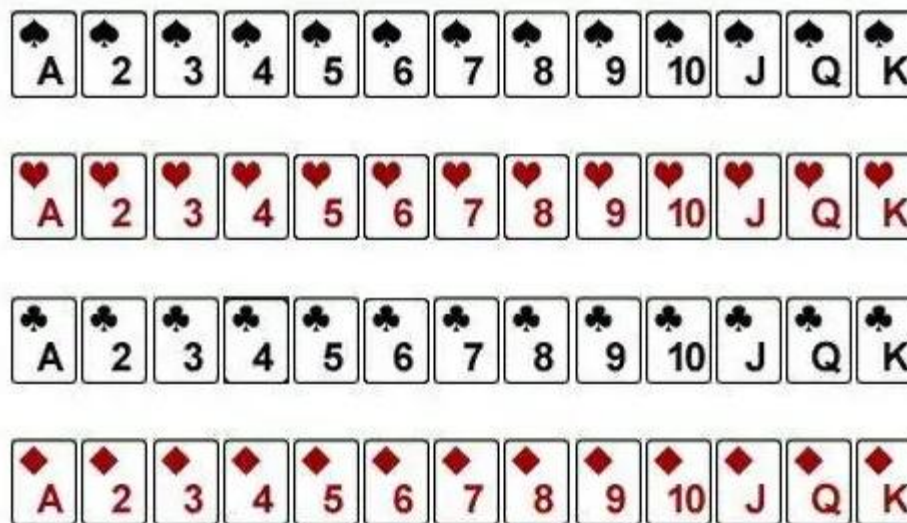
以斗地主为例，为3个玩家分发洗好的牌，展示玩家的牌和3张底牌。

模拟洗牌发牌概述

每副牌有4种花色，每种花色从A-K，另外还有大小王。



"diamonds" 方块
"clubs" 草花
"hearts" 红桃
"spades" 黑桃



生成54张牌

```
#include <iostream>
#include <cstring>
#include <cstdlib>
#include <ctime>
using namespace std;
struct CARD {
    char suit[10]; //花色
    char face[10]; //牌面
} card[54];

const char* suit[] = { "diamonds", "clubs", "hearts", "spades" };
const char* face[] = { "A", "2", "3", "4", "5", "6", "7", "8", "9", "10", "J", "Q", "K" };
void createCards(); //产生54张牌
void shuffle(CARD* card); //洗牌
void dealCards(CARD* card); //发牌
```

```
void createCards() {
    cout << "开始生成54张牌>>>" << endl;
    int i = 0;
    for (; i < 52; i++) {
        strcpy_s(card[i].suit, suit[i / 13]);
        strcpy_s(card[i].face, face[i % 13]);
    }
    strcpy_s(card[52].suit, "red");
    strcpy_s(card[52].face, "joker");
    strcpy_s(card[53].suit, "black");
    strcpy_s(card[53].face, "joker");
    cout << "生成54张牌结束>>>" << endl;
}
```

洗牌

```
void shuffle(CARD* card) {  
    cout << "洗牌开始>>>" << endl;  
    int i = 0, j;  
    CARD temp;  
    srand(time(0));  
    for (; i < 54; i++) {  
        j = rand() % 54;  
        temp = card[i];  
        card[i] = card[j];  
        card[j] = temp;  
    }  
    cout << "洗牌结束>>>" << endl;  
}
```

模拟发牌

```
void dealCards(CARD* card) {
    CARD player[3][17], finalCards[3];
    int p1 = 0, p2 = 0, p3 = 0;
    for (int i = 0; i < 51; i++) {
        if (i % 3 == 0) {
            player[0][p1++] = card[i];
        }
        else if (i % 3 == 1) {
            player[1][p2++] = card[i];
        }
        else {
            player[2][p3++] = card[i];
        }
    }
    for (int i = 51, j = 0; i < 54; i++, j++) {
        finalCards[j] = card[i];
    }

    cout << "开始发牌>>>" << endl;
    for (int i = 0; i < 3; i++) {
        cout << "玩家" << i + 1 << endl;
        for (int j = 0; j < 17; j++) {
            cout << player[i][j].suit << " " << player[i][j].face << endl;
        }
    }
    cout << "底牌" << endl;
    for (int i = 0; i < 3; i++) {
        cout << finalCards[i].suit << " " << finalCards[i].face << endl;
    }
    cout << "发牌结束>>>" << endl;
```

知识点总结

1. 模拟洗牌发牌概述
2. 生成54张牌
3. 洗牌
4. 模拟发牌

作业

模拟自己熟悉的玩法，把生成牌，洗牌，发牌通过指针的方式实现

下节课预告

面向对象